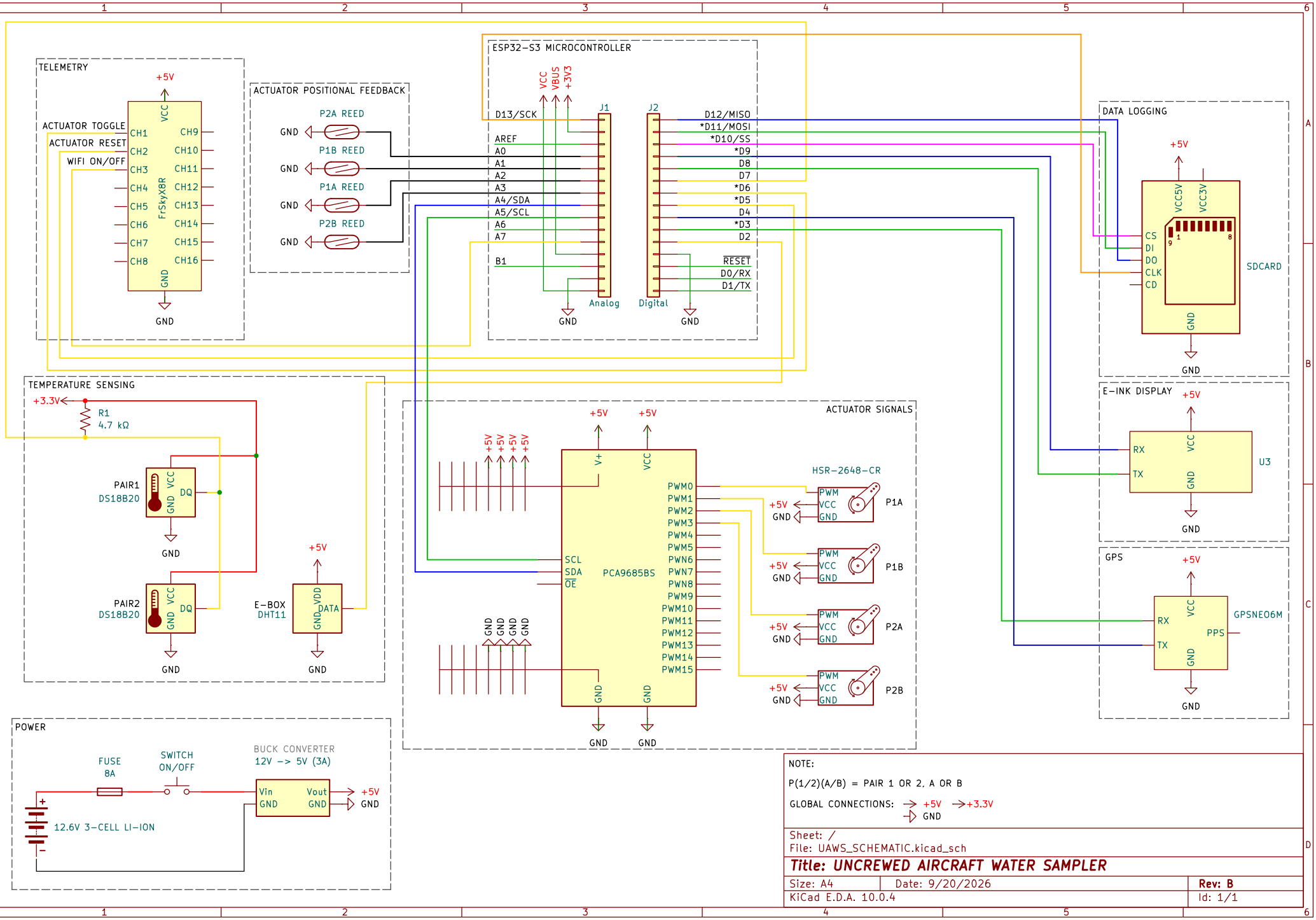
TELEMETRY
+5V
VCC
ACTUATOR TOGGLE
ACTUATOR RESET
WIFI ON/OFF
CH1
CH2
CH3
CH4
CH5
CH6
CH7
CH8
FrSkyX8R
CH9
CH10
CH11
CH12
CH13
CH14
CH15
CH16
GND
GND
ACTUATOR POSITIONAL FEEDBACK
P2A REED
GND
P1B REED
GND
P1A REED
GND
P2B REED
GND
ESP32-S3 MICROCONTROLLER
VCC
VBUS
+3V3
D13/SCK
AREF
A0
A1
A2
A3
A4/SDA
A5/SCL
A6
A7
B1
J1
Analog
GND
J2
D12/MISO
*D11/MOSI
*D10/SS
*D9
D8
D7
*D6
*D5
D4
*D3
D2
RESET
D0/RX
D1/TX
Digital
GND
DATA LOGGING
+5V
VCC5V
VCC3V
CS
DI
DO
CLK
CD
SDCARD
GND
GND
E-INK DISPLAY
+5V
RX
TX
VCC
GND
U3
GND
GPS
+5V
RX
TX
VCC
GND
PPS
GPSNEO6M
GND
TEMPERATURE SENSING
+3.3V
R1
4.7 kΩ
PAIR1
DS18B20
VCC
GND
DQ
GND
PAIR2
DS18B20
VCC
GND
DQ
GND
E-BOX
DHT11
+5V
VDD
GND
DATA
GND
ACTUATOR SIGNALS
+5V
+5V
+5V
+5V
+5V
+5V
V+
VCC
SCL
SDA
OE
PCA9685BS
PWM0
PWM1
PWM2
PWM3
PWM4
PWM5
PWN6
PWN7
PWN8
PWM9
PWM10
PWM11
PWM12
PWM13
PWM14
PWM15
GND
GND
GND
GND
GND
GND
HSR-2648-CR
PWM
VCC
GND
+5V
P1A
P1B
P2A
P2B
POWER
FUSE
8A
SWITCH
ON/OFF
BUCK CONVERTER
12V -> 5V (3A)
Vin
GND
Vout
GND
+5V
GND
12.6V 3-CELL LI-ION
NOTE:
P(1/2)(A/B) = PAIR 1 OR 2, A OR B
GLOBAL CONNECTIONS: +5V +3.3V GND
Sheet: /
File: UAWS_SCHEMATIC.kicad_sch
Title: UNCREWED AIRCRAFT WATER SAMPLER
Size: A4
Date: 9/20/2026
Rev: B
KiCad E.D.A. 10.0.4
Id: 1/1